Date

MID - Term Exam

$\frac{98}{100}$ Very Good!

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(10) 1) $r = a\sin\theta$

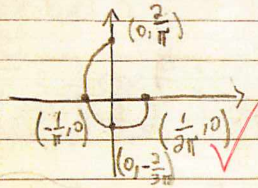
$r^2 = ar\sin\theta \qquad x^2 + y^2 = ay$

$x^2 + \left(y - \frac{a}{2}\right)^2 = \frac{a^2}{4}$ ✓

center : $\left(0, \frac{a}{2}\right)$ ✓

radius : $\frac{a}{2}$ ✓

(18) 2)



$r = \frac{1}{\theta} \qquad x = \frac{1}{\theta}\cos\theta \qquad y = \frac{1}{\theta}\sin\theta$

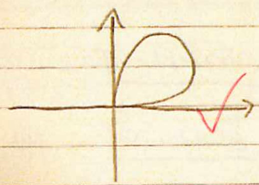
$L = \int_{\frac{\pi}{2}}^{2\pi} \sqrt{\frac{1}{\theta^2} + \frac{1}{\theta^4}}\, d\theta$ ✓

$= \int_{\frac{\pi}{2}}^{2\pi} \sqrt{\frac{\theta^2+1}{\theta^4}} = \int_{\frac{\pi}{2}}^{2\pi} \frac{\sqrt{\theta^2+1}}{\theta^2}$

$= \left[\ln\left(\sqrt{\theta^2+1} + \theta\right) - \frac{\sqrt{\theta^2+1}}{\theta}\right]_{\frac{\pi}{2}}^{2\pi}$ (?) (−2)

≈ 1.476 // ✓

(10) 3) $r = \sqrt{\sin(2\theta)}$



$A = \int_0^{\frac{\pi}{2}} \frac{1}{2}\left[\sqrt{\sin 2\theta}\right]^2 d\theta$ ✓

$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin 2\theta\, d\theta$

$= \frac{1}{2}\left[\frac{-\cos(2\theta)}{2}\right]_0^{\frac{\pi}{2}} = \frac{1}{2} \cdot 1 = \frac{1}{2}$ // ✓

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(10) 4) $ds = \sqrt{dr \cdot dr} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$

$r(t) = e^{-t}\cos t\, \hat{i} + e^{-t}\sin t\, \hat{j} + a\,\hat{k}$

$r'(t) = \left(-e^{-t}\sin t - e^{-t}\cos t\right)\hat{i} + \left(e^{-t}\cos t - e^{-t}\sin t\right)\hat{j} + 0\,\hat{k}$

$|r'(t)| = \sqrt{2\left(e^{-t}\sin(t)\right)^2 + 2\left(e^{-t}\cos(t)\right)^2}$

$L = \int_0^{2\pi} \sqrt{2 \cdot e^{-2t}}\, dt$ ✓ $= \int_0^{2\pi} \sqrt{2}\, e^{-t} dt$

$= \sqrt{2}\left[-e^{-t}\right]_0^{2\pi} = \sqrt{2}\left(-e^{-2\pi} + 1\right)$ // ✓

(20) 5) $K = \frac{1}{R_c} = \frac{|f''(x)|}{\left[1 + (f'(x))^2\right]^{\frac{3}{2}}} \rightarrow R_c = \frac{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{3}{2}}}{\left|\frac{d^2y}{dx^2}\right|}$

$y = e^x$

$R_c = \frac{\left(1 + e^{2x}\right)^{\frac{3}{2}}}{|e^x|}$ ✓

$R_c'(x) = \frac{\frac{3}{2}\left(1+e^{2x}\right)^{\frac{1}{2}}\left(2e^{2x}\right)e^x - e^x\left(1+e^{2x}\right)^{\frac{3}{2}}}{e^{2x}} = 0$

$= \frac{3}{2}\left(2e^{2x}\right) - \left(1 + e^{2x}\right)$

$= 3e^{2x} - 1 - e^{2x} = 0$

$e^x = \pm\sqrt{\frac{1}{2}} \qquad x = \ln\sqrt{\frac{1}{2}}$ ✓ $\qquad R_c\left(\ln\sqrt{\frac{1}{2}}\right) = \frac{\left(1+\frac{1}{2}\right)^{\frac{3}{2}}}{\sqrt{\frac{1}{2}}}$

R_c at smallest is $\frac{\left(1+\frac{1}{2}\right)^{\frac{3}{2}}}{\sqrt{\frac{1}{2}}}$ ✓

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6a) $r(t) = a(1-\cos t)\,\hat{i} + a\sin t\,\hat{j}$

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10 $|r'(t)| = \sqrt{[a(1-\cos t)]^2 + (a\sin t)^2} = \sqrt{a^2 - 2a^2\cos t + a^2\cos^2 t + a^2\sin^2 t}$

$= \sqrt{2a^2(1-\cos t)}$ Knowing that $\cos 2x = 1 - 2\sin^2 x$

$= 2a\sin\frac{t}{2}$

$\mathbb{T}(t) = \frac{r'(t)}{|r'(t)|} = \frac{a(1-\cos t)}{2a\sin\frac{t}{2}}\hat{i} + \frac{a\sin t}{2a\sin\frac{t}{2}}\hat{j}$

$= \sin\frac{t}{2}\hat{i} + \cos\frac{t}{2}\hat{j}$ //

b) $\frac{d\mathbb{T}}{dt} = \frac{1}{2}\cos\frac{t}{2}\hat{i} - \frac{1}{2}\sin\frac{t}{2}\hat{j}$

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$\left|\frac{d\mathbb{T}}{dt}\right| = \frac{1}{2} \Rightarrow \mathbb{N} = \frac{\frac{d\mathbb{T}}{dt}}{\left|\frac{d\mathbb{T}}{dt}\right|} = \cos\frac{t}{2}\hat{i} - \sin\frac{t}{2}\hat{j}$

$\mathbb{B} = \mathbb{T}\times\mathbb{N} = \left(\sin\frac{t}{2}\hat{i} + \cos\frac{t}{2}\hat{j}\right)\times\left(\cos\frac{t}{2}\hat{i} - \sin\frac{t}{2}\hat{j}\right)$

$= -\sin^2\frac{t}{2}\hat{k} + \cos^2\frac{t}{2}(-\hat{k}) = -\hat{k}$ //

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c) $|r'(t)| = 2a\sin\frac{t}{2} \Rightarrow ds = 2a\sin\frac{t}{2}\,dt$

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$S(t) = 4a(-\cos\frac{t}{2}) = -4a\cos\frac{t}{2} + C$

Since its from the origin, $t=0$

$S(0) = -4a + C = 0 \Rightarrow C = 4a$

$\therefore S(t) = 4a(-\cos\frac{t}{2}) + 4a = 4a(1-\cos\frac{t}{2})$ //